Algebra 1: Polynomials

Challenge Level

Name: ____________________

Date: ____________________

Question 1: Simplify the expression $x^2 + 3x + 2$ by factoring.

A) $x(x + 1)$
B) $x(x + 2)$
C) $x^2(x + 3)$
D) $(x + 1)(x + 2)$
E) $(x + 2)(x + 3)$

Question 2: What is the degree of the polynomial $3x^4 - 2x^2 + 1$?

A) 2
B) 3
C) 4
D) 5
E) 6

Question 3: Factor out the greatest common factor from the expression $6x^3 + 12x^2$.

A) $6x^2(x + 2)$
B) $6x^2(x + 1)$
C) $3x^2(2x + 4)$
D) $6x^2(x + 3)$
E) $6x(2x + 4)$

Question 4: Find the value of x for which the expression $x^2 - 4x - 3$ equals zero.

A) $x = 3$
B) $x = -1$
C) $x = 2$
D) $x = -3$
E) $x = 1$

Question 5: Simplify the expression $(x + 1)(x - 3)$ using the distributive property.

A) $x^2 - 2x - 3$
B) $x^2 + 2x - 3$
C) $x^2 - 2x + 3$
D) $x^2 - 3x - 1$
E) $x^2 + 3x - 1$

Question 6: If $f(x) = x^2 + 2x - 3$, find $f(-2)$.

A) -3
B) -1
C) 1
D) 3
E) 5

Question 7: What is the equation of the axis of symmetry for the graph of $y = x^2 - 4x - 3$?

A) $x = -1$
B) $x = 2$
C) $x = 3$
D) $x = -3$
E) $x = 1$

Question 8: Factor the expression $x^3 - 2x^2 - 5x + 6$.

A) $(x - 2)(x - 3)(x + 1)$
B) $(x - 1)(x - 2)(x + 3)$
C) $(x - 2)(x + 1)(x - 3)$
D) $(x - 1)(x + 2)(x - 3)$
E) $(x + 1)(x - 2)(x + 3)$

Question 9: Prove that the polynomial $x^2 + 3x + 2$ can be factored as $(x + 1)(x + 2)$. Show all steps and justify your answer.

Question 10: Find the roots of the polynomial $x^2 - 5x + 6$ using factoring and the quadratic formula. Compare and contrast the two methods.

Chef's Tips (Hints):

- When factoring polynomials, look for the greatest common factor first.
- To find the roots of a polynomial, use factoring or the quadratic formula.
- When using the quadratic formula, make sure to simplify the expression under the square root before solving for x.